



STUDY PROTOCOL

Cross-Cultural Adaptation and Validation of the (Re)-emerging and ePidemic Infectious Diseases Stigma Scales in Thailand: A Study Protocol

[version 1; peer review: 1 approved]

Phyu Hnin Hlaing^{1,2}, Phaik Yeong Cheah^{2,3} , Tassawan Poomchaichote²,
Natinee Kulpijit², Soe Thandar Myint⁴ , Mavuto Mukaka^{2,3} , Amy Paterson⁵,
Sakda Pannengpetch¹, Wattana Viriyasitavat⁶, Smith Boonchutima^{1,7} 

¹Chulalongkorn University Faculty of Communication Arts, Bangkok, Bangkok, 10330, Thailand²Mahidol Oxford Tropical Medicine Research Unit, Bangkok, Bangkok, 10400, Thailand³Nuffield Department of Medicine, University of Oxford Centre for Tropical Medicine and Global Health, Oxford, England, OX1 2JD, UK⁴Mahidol University Faculty of Public Health, Bangkok, Bangkok, 10400, Thailand⁵Pandemic Science Institute, University of Oxford, Oxford, OX3 7DQ, UK⁶Department of Statistics, Chulalongkorn University Faculty of Commerce and Accountancy, Bangkok, Bangkok, 10330, Thailand⁷Center of Excellence in Communication Innovation for Development of Quality of Life and Sustainability, Chulalongkorn University Faculty of Communication Arts, Bangkok, Bangkok, 10330, Thailand

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Abstract

Background

Stigma associated with infectious diseases prevents people from seeking healthcare and undermines outbreak control efforts. This problem affects marginalized populations most severely because they face multiple forms of discrimination during health emergencies. Thailand lacks validated instruments to measure how vulnerable groups experience stigma during outbreaks. This study will adapt and validate the (Re)-emerging and ePidemic Infectious Diseases (RAPID) Community and Self Stigma Scales in two communities in Thailand: (1) mpox-related stigma among Thai men who have sex with men (MSM) and (2) COVID-19 related stigma among Myanmar migrants.

Methods

The study has two phases. Phase 1 takes six months and follows the International Society for Pharmacoeconomics and Outcomes Research

Open Peer Review

Approval Status

1

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[view](#)

1. **Anthony Gifford** , Birmingham City University, Birmingham, UK

Any reports and responses or comments on the article can be found at the end of the article.

(ISPOR) guidelines for cultural adaptation and translation. Professional translators will translate the RAPID scales into Thai and Burmese using forward-backward methods. Eight experts per community will complete three rounds of Delphi evaluation. Cognitive interviews with 15 to 20 members of each community will refine the instruments. Phase 2 requires separate ethics approval and will validate the adapted scales with 400 participants (200 per community). Validation will include confirmatory factor analysis, reliability testing, and validity assessment.

Expected Outcomes

This research will create the first validated tools for measuring how Thai MSM communities and Myanmar migrants experience stigma during infectious disease outbreaks in Southeast Asia. The Thai and Burmese versions of the RAPID scales can be deployed quickly during future outbreaks to monitor stigma in real time. The instruments will help health systems identify barriers to care and design interventions that address structural, interpersonal, and internalized forms of stigma. These tools may prove useful throughout Thailand and other countries in the region.

Plain Language summary

The Stigma Scales are newly validated questionnaires developed by Oxford University to measure stigma during infectious disease outbreaks. They have been proven to work across different diseases (Ebola, mpox, Nipah virus) and different countries, making them the first questionnaires of their kind that can be used across different outbreaks.

This is a cultural adaptation and translation study - we're making sure the scales work properly in Thai and Burmese languages and cultural contexts.

We're focusing on two vulnerable populations in Thailand:

- Thai men who have sex with men (MSM)-to measure stigma related to mpox

Each community will have 200 people taking part in the study.

Phase 1: Translating the scales into Thai and Burmese using professional translation methods, adapting the questions for cultural appropriateness, and interviewing people from both Thai MSM and Myanmar migrant communities to make sure the questions make sense to them.

Phase 2: Collecting questionnaire response from 400 people and using statistical methods to check that the scales give reliable and meaningful results in Thai and Burmese.

This will give Thailand and Southeast Asia the first validated tools to quickly assess stigma during outbreaks, enabling real-time intervention design.

Keywords

stigma measurement, cross-cultural adaptation, psychometric validation, infectious disease, men who have sex with men, Myanmar migrants, Thailand, RAPID scales, mental health, ethics

This article is included in the [Mahidol Oxford](#)



[Tropical Medicine Research Unit \(MORU\)](#) gateway.

Corresponding author: Smith Boonchutima (Smith.b@chula.ac.th)

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Introduction

Stigma represents one of the most persistent obstacles to effective infectious disease control worldwide.¹ When communities stigmatize individuals affected by infectious diseases, the consequences extend beyond individual suffering to compromise public health responses.¹ Stigmatized individuals frequently delay or avoid seeking medical care, conceal symptoms from healthcare providers and family members,² and resist participation in contact tracing and outbreak investigations.³ These stigma-driven behaviors directly undermine outbreak control efforts by facilitating disease transmission while perpetuating cycles of fear and discrimination.¹

Recent outbreaks have repeatedly demonstrated the profound impact of stigma on health-seeking behavior. During the COVID-19 pandemic, stigma toward certain ethnic groups and occupations significantly impeded testing and vaccination efforts across multiple countries.⁴ In Thailand, Nochaiwong and colleagues developed the COVID-19 Public Stigma Scale (COVID-PSS) to assess public attitudes toward COVID-19 infection, demonstrating that stigma significantly predicted both symptom-reporting hesitancy and care-seeking delays among the general population.⁵ Globally, the 2022-2023 mpox outbreak revealed how stigma toward men who have sex with men (MSM) created substantial barriers to healthcare access, with many individuals delaying diagnosis and treatment due to fears of discrimination and unwanted disclosure of sexual orientation.⁶

Despite growing recognition of stigma's critical role in outbreak response, significant measurement gaps remain. While disease-specific instruments like the COVID-PSS provide valuable snapshots of public attitudes toward particular infections, they require substantial development time and resources for each new outbreak context.⁵ More critically, existing instruments predominantly capture general population perceptions rather than the lived experiences of stigma among vulnerable populations who face compounded discrimination during outbreaks. Traditional approaches involve developing disease-specific scales, a process requiring approximately two years from outbreak onset to scale publication.⁴ This timeline renders most stigma assessment tools essentially useless for informing interventions during the active phase of an outbreak, when stigma tends to be most severe and its public health consequences most acute.

The (Re)-emerging and ePidemic Infectious Diseases (RAPID) Stigma Scales, developed by the University of Oxford's research team, represent a significant methodological advancement.⁷ Unlike disease-specific instruments that assess public attitudes toward infected individuals, the RAPID scales were designed to measure both community-level stigma manifestations and internalized self-stigma among affected populations themselves, a critical distinction for understanding healthcare barriers among marginalized groups.⁷ These scales were explicitly designed for transferability across different outbreak contexts, enabling rapid deployment without extensive redevelopment. The scales underwent rigorous validation across three diverse outbreak contexts: the Ebola disease in Uganda, mpox in the United Kingdom, and Nipah virus disease in Bangladesh with a total of 1,008 participants. The validation exercise demonstrated robust psychometric properties with ordinal alpha reliability coefficients ranging from 0.79 to 0.92.⁷

Thailand currently lacks validated instruments specifically designed to assess stigma and experiences among vulnerable populations during infectious disease outbreaks.⁸ Existing stigma measurement tools in Thai contexts have not been designed for cross-outbreak application or validated with marginalized communities facing intersectional discrimination during health emergencies. Thai MSM communities and Myanmar migrants were selected because they represent two distinct stigma mechanisms in the Thai context: Thai MSM face compounded identity-based and disease-related discrimination during mpox outbreaks,⁶ while Myanmar migrants experience intersectional stigma arising from both migration status and infection-related discrimination during the COVID-19 pandemic.⁹ The adaptation and validation of the RAPID Stigma Scales for these two communities addresses this gap by providing culturally appropriate, rapidly deployable tools that capture the lived experiences of stigma among those most affected by discrimination related healthcare barriers^{10–12}

Study objectives and research questions

Primary objectives

Phase 1 (Cultural adaptation and Translation):

1. Conduct systematic translation and cultural adaptation of the RAPID Community and Self Stigma Scales from English to Thai (mpox context) and Burmese (COVID-19 context) following ISPOR guidelines
2. Establish content validity through expert panel evaluation and cognitive interviews with both target populations
3. Produce culturally appropriate draft scales ready for psychometric validation

Phase 2 (Psychometric Validation):

1. Establish construct validity through confirmatory factor analysis
2. Assess internal consistency reliability
3. Evaluate external construct validity through hypothesis testing
4. Develop implementation guidelines for regional health agencies

Research questions**Phase 1:**

1. How can the RAPID Stigma Scale items be linguistically translated and culturally adapted to accurately capture mpox-related stigma among Thai MSM communities and COVID-19 related stigma among Myanmar migrants?
2. What cultural modifications are necessary to ensure items are comprehensible, relevant, and appropriate for both populations and disease contexts?
3. Do the adapted scale items demonstrate adequate content validity as assessed by expert panels and target population representatives?

Phase 2:

1. Do the adapted scales demonstrate acceptable structural validity as evidenced by confirmatory factor analysis?
2. What level of internal consistency reliability do the adapted scales achieve?
3. To what extent do scale scores predict theoretically expected outcomes including symptom-reporting hesitancy, care-seeking behavior, and social acceptance of recovered individuals?

Methodology**Study design overview**

This research employs a two-phase sequential design. Phase 1 focuses exclusively on cultural adaptation and translation over 5 months, involving systematic forward-backward translation, expert Delphi panel evaluation, and cognitive interviews with target populations. Phase 2 will involve full psychometric validation through survey administration to 400 participants over 7 months. The study follows the Boateng framework for cross-cultural scale adaptation¹³ while maintaining close collaboration with research team at Oxford University who developed the original RAPID Stigma Scales⁷ to ensure adaptations remain faithful to theoretical foundations.

The study operates through a dual-arm parallel implementation structure, with Group 1 focusing on Thai MSM communities and Arm 2 focusing on Myanmar migrants. This structure enables efficient resource use, facilitates real-time cross-population learning, and helps identify universal versus population-specific adaptation needs.

This adaptation involves two distinct disease contexts reflecting documented outbreak experiences among vulnerable populations in Thailand:

Group 1- Thai MSM Communities (Mpox-Related Stigma)

The adaptation focuses on mpox-related stigma, addressing the documented healthcare access barriers and discrimination experienced by Thai MSM during the 2022-2023 mpox outbreak. Mpox stigma among Thai MSM communities reflects both disease-related and identity-based discrimination, creating compound barriers to testing, treatment, and care-seeking.

Group 2 – Myanmar Migrants (COVID-19-Related Stigma)

The adaptation focuses on COVID-19 related stigma, addressing the well-documented barriers Myanmar migrants faced during the pandemic including language barriers, fear of deportation, employment insecurity, and discrimination in accessing healthcare services. COVID-19 stigma among Myanmar migrants reflects intersectional vulnerabilities related to both migration status and infection related discrimination.

This dual- context approach will produce two culturally and linguistically adapted versions:

- Thai language RAPID scales (mpox context) for Thai MSM communities
- Burmese language RAPID scales (COVID-19 context) for Myanmar migrants

Both versions maintain the original RAPID scales' theoretical framework and item structure while incorporating disease-specific terminology and culturally appropriate adaptations identified through expert panels and cognitive interviews.

Phase 1: Cultural adaptation and translation

Item Phrasing Approach

The adapted scales will maintain the original RAPID scales' third-person/distanced phrasing (e.g., “People who have mpox are looked down on” rather than “I am looked down on”). This distanced approach reduces participant burden, enables broader participation beyond those with lived experience, and aligns with the validated Oxford methodology. During cognitive interviews, we will assess whether this phrasing adequately captures stigma experiences in each population, with flexibility to modify specific items in consultation with the Oxford team if cultural factors warrant direct experience phrasing.

Translation Process (International Society for Pharmacoeconomics and Outcomes Research (ISPOR) guidelines)

The translation follows systematic procedures recommended by ISPOR through five sequential steps (Figure 1).¹⁴ Two independent professional translators for each language produce complete forward translations. For Thai translation, one translator will be a healthcare professional with experience in sexual health and LGBTQI+ populations, while the second

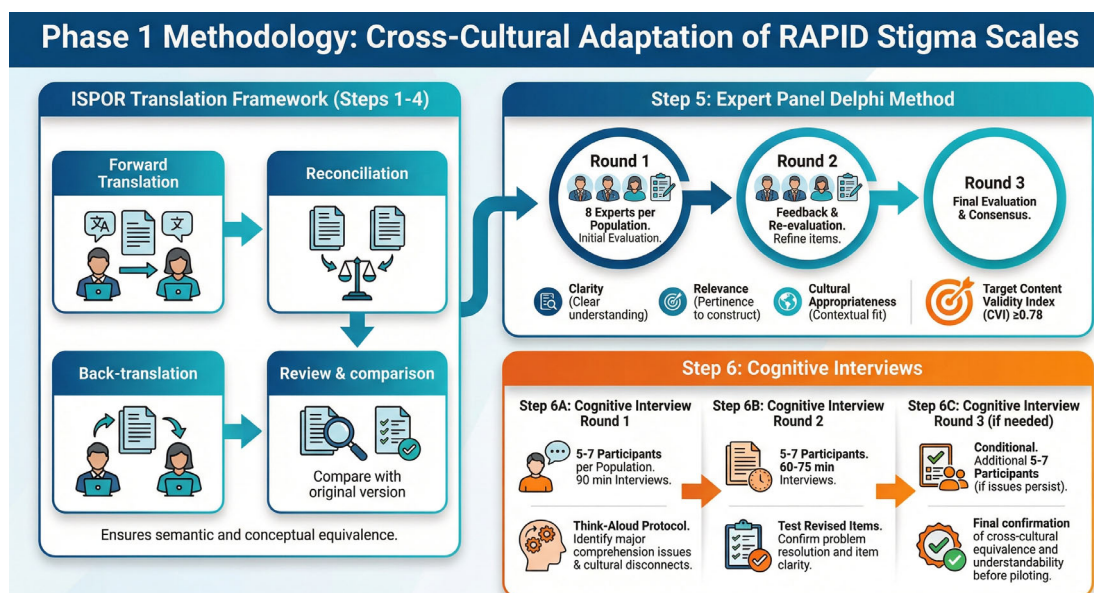


Figure 1. Phase I methodology: Cross-cultural adaptation of RAPID stigma scales. Systematic cross-cultural adaptation process following ISPOR translation guidelines. Steps 1-4 (left) show the translation framework including forward translation, reconciliation, back-translation, and review to ensure equivalence with original scales. Steps 5-6 (right) show the validation process through three rounds of expert Delphi panel assessment and cognitive interviews with Thai MSM communities and Myanmar migrants to ensure cultural appropriateness and comprehension.

will be a social scientist with expertise in sexual minority health. For Burmese translation, one translator will be a healthcare professional with migrant health experience, while the second will be a social scientist specializing in migration health with familiarity with Myanmar ethnic diversity.

Following forward translation, separate reconciliation panels for each language produce single, synthesized translations. The Thai panel includes both forward translators, a bilingual research team member with expertise in Thai health contexts, and a Thai MSM community representative. The Burmese panel similarly includes both forward translators, a bilingual research team member with Myanmar health expertise, and a Myanmar community representative familiar with migrant healthcare experiences.

Each reconciled translation then undergoes back-translation into English by two different professional translators who were not involved in forward translation and lack specialized knowledge of stigma research or the RAPID scales. This intentional lack of familiarity provides objective assessment if back-translators unfamiliar with the original instrument can produce English translations closely matching the original, this provides strong evidence that translations successfully capture intended meaning.

Expert Panel Evaluation (Delphi Methodology)

Two separate expert panels (one per community), each comprising eight carefully selected members, will evaluate the translated scales through three structured Delphi rounds. The Thai MSM panel includes 2 LGBTQI+ community leaders with lived experience, 2 sexual health specialists with clinical experience, 2 public health researchers with MSM health expertise, 1 Thai language expert, and 1 cultural/religious studies expert. The Myanmar migrant panel includes 2 Myanmar community leaders with migration experience, 2 Thai healthcare professionals experienced with migrant populations, 2 migration health researchers, 1 Burmese language expert, and 1 Myanmar cultural diversity expert.

Round 1 involves independent evaluation where experts rate each item on three dimensions using four-point Likert scales: clarity (1 = Not clear to 4 = Highly clear), relevance (1 = Not relevant to 4 = Highly relevant), and cultural appropriateness (1 = Not appropriate to 4 = Highly appropriate).¹⁵ Between rounds, the research team conducts comprehensive analysis calculating Item-level Content Validity Index (I-CVI), Scale-level CVI (S-CVI), Content Validity Ratio (CVR), and Cohen's kappa for inter-rater agreement¹⁶ may be considered. Items with I-CVI below 0.78 are revised to address identified concerns.

Round 2 evaluates only revised items, with experts indicating whether revisions successfully address Round 1 concerns. Round 3 serves as final consensus evaluation where experts assess the entire adapted scale in near-final form, providing final ratings and responding to summative questions about comprehensiveness, appropriateness for outbreak use, and readiness for cognitive interviewing.

Cognitive Interviews with Target Populations

Cognitive interviews with 15-20 participants per population will verify items are comprehensible,¹⁷ interpreted as intended, culturally appropriate, and that response options are adequate. The interviews proceed through 2-3 iterative rounds with 5-7 participants each, using think-aloud techniques with targeted probing guided by comprehension, response process, and cultural appropriateness questions.¹⁸

For Thai MSM, purposive sampling ensures diversity across age (18-30, 31-45, 46+ years), sexual health service engagement levels, geographic distribution, education levels, and HIV status. Recruitment occurs through LGBTQI+ community organizations, sexual health clinics, social media forums, dating applications, and peer referral. For Myanmar migrants, sampling ensures diversity across documentation status, ethnic groups (Bamar, Karen, Shan, Mon), geographic distribution, gender, and length of residence in Thailand. Recruitment occurs through community-based organizations, religious institutions, healthcare facilities, workplace partnerships, and village health volunteers.

Trained bilingual research staff conduct 60-90 minute interviews in culturally appropriate settings.¹⁹ The semi-structured protocol systematically addresses each scale item, with participants reading items aloud, explaining their understanding, identifying confusing elements, and discussing relevance to their experiences. Iterative refinement continues until saturation is achieved, defined as 3-4 consecutive interviews yielding no new comprehension issues, cultural concerns, or suggestions for improvement.

Phase 2: Psychometric validation

Overview

Phase 2 involves psychometric validation through cross-sectional survey administration (Figure 2)¹³ to 400 participants (200 per population). This sample size of 200 per population is consistent with Boateng GO (2018; Guadagnoli E and Velicer WF (1988) and Comrey AL (1988) that indicate that range of 200–300 is appropriate for factor analysis.^{13,20,21} The RAPID scales contain 16 items (12 community stigma, 4 self stigma), making this sample adequate for all planned analyses. A separate ethical review is required for this phase.

Sampling and Recruitment

Non-random quota sampling will ensure representation across five subgroups per population: recovered individuals (20-30), close contacts (40-50), healthcare workers (40-50), outbreak support staff (20-30), and general community members (50-70). This approach addresses the risk that probability sampling would under-represent affected populations while managing logistical challenges in resource-limited settings.

Thai MSM recruitment will occur through LGBTQI+ organizations, sexual health clinics, HIV testing services, social media, dating apps, and peer networks, ensuring diversity across age, service engagement, geography, education, and HIV status. Myanmar migrant recruitment will occur through community organizations, religious institutions, migrant clinics, workplace partnerships, and community leaders, ensuring diversity across documentation status, ethnicity, geography, gender, and residence duration.

Eligibility: Age ≥ 18 years, residence in Thailand (Thai MSM) or status as Myanmar migrants, awareness of mpox (Thai MSM) or COVID-19 (Myanmar migrants), fluency in Thai/Burmese, and ability to consent. **Exclusions:** Cognitive impairment or acute distress.

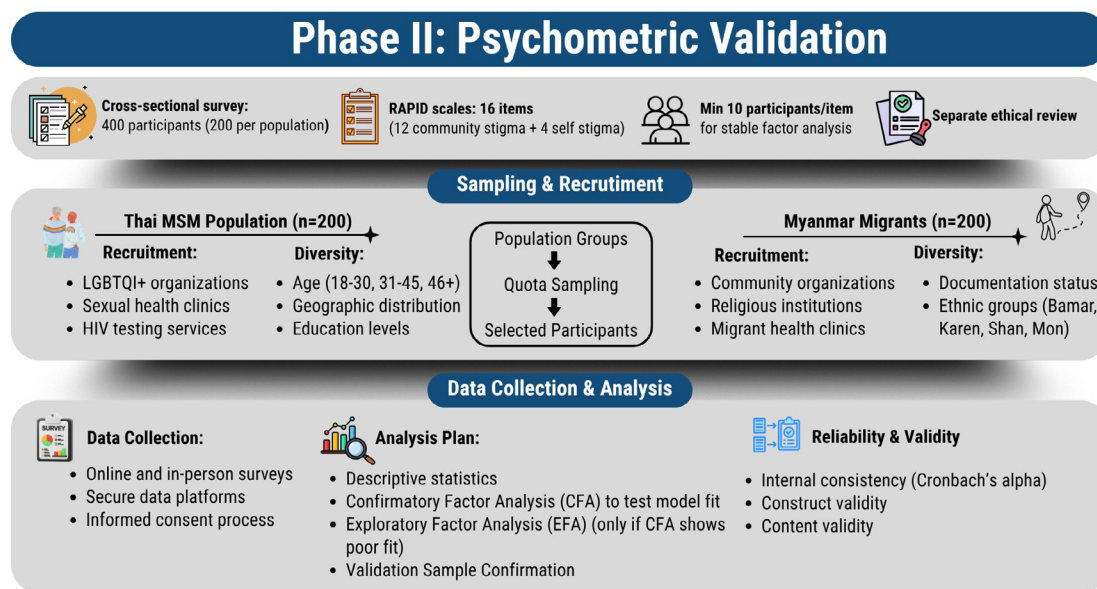


Figure 2. Phase II methodology: Psychometric validation. Psychometric validation study design employing quota sampling to recruit 400 participants: Thai MSM communities (n = 200) recruited through LGBTQ+ organizations and sexual health clinics, and Myanmar migrants (n = 200) recruited through community organizations and migrant health centers. Data collection uses online and in-person surveys in Thai and Burmese. Analysis plan includes descriptive statistics, Confirmatory Factor Analysis (CFA) to test the 4-factor model structure, and reliability and validity assessments (internal consistency, construct validity, content validity) following established psychometric standards.

Survey Administration

Surveys are administered via mobile-optimized REDCap with two embedded attention checks (both must pass), progress tracking, and automatic validation. Paper forms serve as backup in areas with poor internet. The 15-20 minute survey includes consent, eligibility screening, demographics (10-15 items), RAPID Community Stigma Scale (12 items), RAPID Self Stigma Scale (4 items), and external validity items assessing symptom-reporting hesitancy (3 items), care-seeking intentions (2 items), and social acceptance (4 items).

Statistical analysis

All analyses use R (version 4.4.2+) with packages psych, GPArotation, lavaan, semTools, and survey or other comparable analysis software. Statistical significance will be declared at $p < 0.05$.

Data cleaning removes ineligible respondents, incomplete surveys (<80% completion), failed attention checks, and duplicates. Missing data (expected <0.5% based on Oxford findings) is handled via listwise deletion. Sensitivity analysis using multiple imputation or other principled statistical methods if missing will exceed 5%. A detailed Statistical analysis plan will be developed before database lock.

Descriptive and response distribution analysis

Participant characteristics are summarized with standard descriptive statistics stratified by population and quota category. For each item, we calculate response distributions, means, medians, skewness, kurtosis, and floor/ceiling effects. Items are flagged if they show floor/ceiling effects >80%, skewness outside -1 to +1, kurtosis >7, adjacent endorsement <10%, or inter-item correlations >0.8. Continuous variables with skewed distribution will be summarised using medians (interquartile-range) will normally distributed variables will be summarised using mean (standard deviation). Count data will be summarised using frequency counts and percentages.

Factor analysis

We employ a sequential confirmatory -exploratory approach appropriate for cross-cultural adaptation of validated instruments. The dataset splits randomly into equal primary and validation halves (200 each), maintaining population balance and subgroup representation.

Step 1: Confirmatory Factor Analysis (Primary Sample, n = 200)

Confirmatory Factor Analysis (CFA) tests whether the original RAPID factor structure replicates in Southeast Asian contexts, using weighted least squares means and variance adjusted (WLSMV) estimator appropriate for ordinal data. Factorability is first confirmed via Kaiser-Meyer-Olkin measure (>0.8) and Bartlett's test ($p < 0.05$).

For **Community Stigma (12 items)**, we test the original second-order model with four first-order factors:

- Initial Social Stigma (3 items: looked down on, gossiped about, treated unkindly)
- Provider/Authority-Related Stigma (3 items: healthcare worker judgment, media portrayal, political rhetoric)
- Structural Stigma (3 items: denied rights, work/education loss, exclusion from places)
- Enduring Social Stigma (3 items: partnership difficulties, business impact, community rejection)

For **Self Stigma (4 items)**, we test the original unidimensional model (secrecy, shame, care-seeking hesitancy, self-blame).

Model fit criteria:

- Good fit: CFI >0.95, RMSEA <0.06, SRMR <0.08
- Acceptable fit: CFI >0.90, RMSEA <0.08, SRMR <0.10
- Factor loadings: Target ≥ 0.6 (good) or ≥ 0.4 (acceptable)
- Local fit: Residual correlations <0.10, modification indices <15

CFI: Comparative Fit Index

RMSEA: Root Mean Square Error of Approximation

SRMR: Standardized Root Mean Square Residual

Decision Rules:

- If CFA shows good/acceptable fit → Structure confirmed, proceed to Step 3 (validation)
- If CFA shows poor fit (CFI < 0.90 or RMSEA > 0.08) → Proceed to Step 2 (EFA)

Step 2: Exploratory Factor Analysis (EFA) (Conditional - Only if CFA Shows Poor Fit)

If CFA indicates the original structure does not replicate adequately, EFA identifies culturally appropriate factor structures while maintaining theoretical grounding in health-related stigma frameworks.

EFA uses unweighted least squares extraction and Promax rotation (allowing factor correlation), with optimal factor number determined by:

- Parallel analysis (simulated random data comparison)
- Empirical Kaiser criterion (factors with eigenvalues > 1.0)
- Theoretical interpretability (alignment with stigma theory)

Item retention criteria:

- Primary factor loadings > 0.4
- Cross-loadings < 0.3
- Communalities > 0.4

Items failing to meet criteria are removed iteratively (one at a time, worst-performing first), with EFA rerun after each removal until all remaining items meet criteria. Bootstrap confidence intervals (1,000 resamples with Procrustes rotation) provide precision estimates for factor loadings.

If EFA is conducted, results will be discussed with Oxford research team to assess theoretical validity of any emergent structure. Modified structures will be documented with empirical and theoretical justification, cross-cultural comparability limitations will be explicitly noted in all reporting, and population-specific scoring algorithms may be developed in consultation with the Oxford team to balance cultural appropriateness with international comparability.

Step 3: Validation Sample Confirmation (n = 200)

The structure confirmed or identified in Steps 1-2 is tested in the validation sample using CFA with identical fit criteria. This split-sample validation ensures the factor structure is stable and not an artifact of sample-specific characteristics.

Validation by population

The factor structure is validated separately for each population (n = 200 each) using CFA with identical fit criteria. This approach recognizes that the two populations experience stigma in different disease contexts (mpox vs. COVID-19), making direct score comparisons inappropriate. Separate reliability and validity assessments are conducted for each adapted scale.

Sensitivity analysis: Weighted CFA

To assess the impact of non-random quota sampling, we conduct weighted CFA applying sampling weights based on estimated population sizes for each quota category. Weighted and unweighted models are compared on fit indices and factor loadings. Substantial differences (Δ loadings > 0.1) indicate sampling effects requiring interpretation adjustment.

Reliability

Internal consistency is assessed via ordinal alpha (primary), Cronbach's alpha, and hierarchical omega (secondary), with criteria ≥ 0.70 (acceptable), ≥ 0.80 (good), ≥ 0.90 (excellent).²² Average Variance Extracted (AVE > 0.50) indicates convergent validity.²³ Reliability is calculated for all subscales and overall scales by population.

External construct validity

Four pre-specified hypotheses are tested via regression analyses controlling for age, gender, site, residence, healthcare worker status, previous diagnosis, close relationships with recovered individuals, and disease understanding:

H1: Community and self stigma predict higher symptom-reporting hesitancy (multiple linear regression, expected $\beta > 0$, $p < 0.05$).

H2: Both stigma types predict care-seeking hesitancy (multivariable logistic regression, expected OR > 1.0 , $p < 0.05$).

H3: Community stigma predicts lower social acceptance (multiple linear regression, expected $\beta < 0$ or OR < 1.0 , $p < 0.05$).

H4: Community stigma predicts self stigma based on internalization theory (multiple linear regression, expected $\beta > 0$, $p < 0.05$).

All models check variance inflation factors (VIF < 10) and report R^2 for fit.

Final content validity

Post-CFA, Delphi experts rate face validity (target $> 80\%$ scoring 3-4 on 4-point scale), item relevance (r-CVI > 0.80 per item, S-CVI > 0.90 overall), and comprehensiveness (c-CVI > 0.80).¹⁵

Sample size and power

The sample size of 400 is consistent with Smith et al., 2025¹⁹ recommendation and provides more than 80% power to detect moderate effects ($\beta \approx 0.20$, OR ≈ 1.5) testing at 5 % significance level (i.e. $\alpha = 0.05$). Per-population analyses ($n = 200$) are consistent with Boateng GO (2018); Guadagnoli E and Velicer WF (1988) and Comrey AL (1988) that indicate that a range of 200–300 is appropriate for factor analysis.

Quality assurance and ethics

Research staff will complete a 2-day training covering protocols, cultural competency, interviewing techniques, and REDCap. Quality monitoring includes weekly meetings, random audio checks (10%), real-time data monitoring, and double data entry for paper forms ($> 99\%$ agreement target).

Thai MSM protections: Recruitment through trusted LGBTQI+ organizations, affirming interview settings, culturally competent staff, no HIV status collection (optional self-disclosure), *and clear messaging about sexual orientation recording*.

Myanmar migrant protections: This study has no connection to immigration authorities or any government agency responsible for immigration matters and no information will be shared or made accessible to any such bodies at any stage of the research. No documentation status collection, visa/permit information will be collected, and no identity document will be required to participate. Explicit safety assurances regarding these protections will be provided both verbally and in writing. Recruitment will be endorsed by local community leaders who have reviewed the research procedures in advance.

Distress protocols: Pause/discontinue if needed, provide psychosocial support referrals (Rainbow Sky Association and Service Workers in Group Foundation for Thai MSM; and Migrant Assistance Program Foundation (MAP) for Myanmar migrants).

Data protection: Paper forms in locked cabinets, encrypted REDCap servers, audio recordings destroyed post-transcription, 10-year anonymized storage on institutional servers with restricted access. All data handling procedures comply with Thailand's Personal Data Protection Act B.E.2562 (2019) and the UK Data Protection Act 2018 and UK General Data Protection Regulation (UK GDPR).

Informed Consent: Written or digital informed consent will be obtained from all participants in their preferred language (Thai, Burmese, or English) prior to data collection. The consent process covers study purpose, procedures, voluntary

participation, anonymous data use, withdrawal rights, potential risks (emotional discomfort), and available psychosocial support resources.

Conclusion

This systematic cross-cultural adaptation and validation of the RAPID Stigma Scales addresses a critical gap in Thailand's pandemic preparedness infrastructure while contributing essential knowledge about effective stigma measurement in LMIC contexts. By following rigorous methodological standards and maintaining close collaboration with international experts, this research will produce culturally appropriate, psychometrically sound instruments that enable evidence-based stigma reduction interventions for vulnerable populations facing documented healthcare access barriers. The success of this work will establish a model for similar adaptations throughout Southeast Asia and demonstrate the feasibility and value of investing in cross-outbreak assessment tools as part of comprehensive pandemic preparedness strategies.

Ethics statement

This study will be conducted in strict accordance with ethical standards for research involving human participants. The research protocol has been submitted to the Research Ethics Review Committee for Research Involving Human Research Participants, Group I, Chulalongkorn University, Bangkok, Thailand. Ethics approval is currently pending, with submission reference number-680344.

The study will only commence data collection activities upon receiving formal ethics approval. The approval will be valid for the duration of the research project (18 months), with annual renewal applications submitted as required by the committee.

All research activities will be conducted in accordance with relevant ethical guidelines and regulations, including the Belmont Report (1979), the Declaration of Helsinki (2013), and the International Ethical Guidelines for Health-related Research Involving Humans (CIOMS, 2016).

Use of AI and generative tools

The authors used Claude (Sonnet 4.5, Anthropic) for limited language editing to improve sentence clarity. [Figure 1](#) was created using Genspark AI based on author specifications. [Figure 2](#) was created using Canva design tools. All content was reviewed, edited, and verified by the authors, who take full responsibility for the manuscript.

Figures

All figures and tables included in this submission are original creatin by the authors and have not been previously published or used in any other publication.

Data availability

Underlying data

No data associated with this article.

Extended data

Open Science Framework: Cross-Cultural Adaptation and Validation of RAPID Stigma Scales <https://doi.org/10.17605/OSF.IO/GZM9K>²⁴

This project contains the following underlying data:

Data Collection Tools - Cognitive Interviews:

- Cognitive Interview Guide_English. (English version)
- Cognitive Interview Guide_ThaiMSM_Thai. (Thai language version for Thai MSM community)
- Cognitive Interview Guide_MMmigrant_Burmese. (Burmese language version for Myanmar migrants)

Data Collection Tools – Delphi Panels:

- Delphi Round 1_ThaiMSM_English. (Delphi Round 1 evaluation form for Thai MSM community)
- Delphi Round 1_MMmigrant_English. (Delphi Round 1 evaluation form for Myanmar migrants)

- Delphi Round 2_English. (Delphi Round 2 evaluation form)
- Delphi Final Round_English. (Delphi final round evaluation form)

Participant Information Sheets:

- PIS_Cognitive Interview (Thai MSM_English). (Participant information sheet for cognitive interviews with Thai MSM community, English version)
- PIS_Cognitive Interview (Thai MSM_Thai). (Participant information sheet for cognitive interviews with Thai MSM community, Thai version)
- PIS_Cognitive Interview (Myanmar Migrants_English). (Participant information sheet for cognitive interviews with Myanmar migrants, English version)
- PIS_Cognitive Interview (Myanmar Migrants_Burmese). (Participant information sheet for cognitive interviews with Myanmar migrants, Burmese version)
- PIS_Delphi_(Myanmar Migrants_English). (Participant information sheet for Delphi panel with Myanmar migrant experts, English version)
- PIS_Delphi_(Myanmar Migrants_Burmese). (Participant information sheet for Delphi panel with Myanmar migrant experts, Burmese version)
- PIS_Delphi_ThaiMSM_English. (Participant information sheet for Delphi panel with Thai MSM experts, English version)
- PIS_Delphi_ThaiMSM_Thai. (Participant information sheet for Delphi panel with Thai MSM experts, Thai version)

Reporting guidelines

- COSMIN_Reporting_Guideline_Checklist. (COSMIN Reporting Guideline version 2.0 for studies on measurement properties of patient-reported outcome measures (Gagnier JJ, et al. Quality of Life Research 2025).²⁵

Extended data are available under the terms of the [Creative Commons Attribution 4.0 International license \(CC-BY 4.0\)](#).

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Anthony Gifford 

Birmingham City University, Birmingham, UK

Thank you for allowing me to review this protocol and I believe this to be of great interest, aligning with my own work. In general, the rationale is clearly articulated, and the study demonstrates strong applied relevance for public health preparedness in Southeast Asia. I have some minor points to comment on as follows:

The authors are to be commended for the methodological rigour of the proposed design, particularly the use of ISPOR guidelines and cognitive interviewing to support cross-cultural adaptation. While I have not directly adapted an existing scale, I have experience in the development of psychometric instruments, and I approached this review with particular attention to the proposed validation strategy.

As this is a protocol, scrutiny of the methods is especially important; however, it is also understood that there remains scope for iterative refinement as the study progresses. In that regard, this work provides a strong and thoughtful foundation, and it will be exciting to see how the empirical data ultimately inform the final structure of the scales.

I offer a small number of suggestions aimed at strengthening clarity and justification within the methods section. First, the decision to conduct confirmatory factor analysis (CFA) prior to exploratory factor analysis (EFA) is defensible given the theoretical grounding. However, as this approach differs from more conventional EFA-first sequencing, it would be helpful for the authors to expand their justification to make this rationale more explicit to readers to enhance replication.

Second, while model fit thresholds and factor loading criteria are clearly stated, the inclusion of additional supporting references would strengthen the transparency and defensibility of these analytic decisions. This is not a concern regarding the appropriateness of the thresholds themselves, but rather an opportunity to more clearly anchor them within established psychometric literature.

Finally, there are a small number of minor typographical and formatting issues throughout the

manuscript (e.g., spacing and minor inconsistencies) which could be addressed in revision to further improve readability.

I look forward to seeing the protocol in action!

Is the rationale for, and objectives of, the study clearly described?

Yes

Is the study design appropriate for the research question?

Yes

Are sufficient details of the methods provided to allow replication by others?

Yes

Are the datasets clearly presented in a useable and accessible format?

Not applicable

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Psychology; Psychometrics; Sexual Health; LGBTQ+ Psychology

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.
